

No Distributed Address-Only KV Class in RoPE Decoders

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Abstract

Keys determine *where* attention can go; values determine *what* it retrieves. A tempting inference-cache optimization is to retain some old non-prefix keys but zero their values, saving half the per-entry bytes while preserving the softmax denominator. We preregistered a strict conjunction test for whether such a *distributed, byte-substantial* address-only class exists and built an optimizer-robust audit: exact ternary (DROP, ADDRESS, or FULL) coalition enumeration with grouped-query-attention semantics and W_O -projected error, an additive-surrogate allocator, and multistart nonlinear local search. The search matched exhaustive optimization on all 15,840 audited strategy rows; independent reconstruction differed by at most 4.857×10^{-17} . On regenerated activations from three pinned RoPE decoders (Qwen2.5-3B, Qwen2.5-7B, and Mistral-7B) over three disjoint PG-19 documents at 2K, 4K, and 8K query endpoints, the aggregate finding is negative: the strongest allocator selects only **1.3415%** ADDRESS entries, yielding **0.6708%** corrected byte saving, and every model, document, and endpoint is individually NO-GO against the preregistered 20% ADDRESS and 15% saving gates. Pointwise absence is nevertheless false: 50 gate-passing strategy rows occur across 26 of 440 model/layer/head locations, including a reproducible Mistral layer-0 pocket spanning KV heads 1–4. A preregistered exact-layer-matched sink-overlap permutation test is null (Holm $p=1.0$ for both metrics), so we reject the “sink-adjacent” label and describe sparse early-layer exceptions instead. Powered ± 1 -point RULER equivalence between address-only and binary caches holds for all three models (power 0.928, 0.923, and 0.997). Outside the pockets, drop and address errors co-move, with per-model median ratios near 1.0 over 17.6 million eligible entries. We thus report a bounded negative and release a verifier that recomputes every coalition loss, gate decision,

and statistical test.

1 Introduction

Modern long-context inference is bottlenecked by the key–value (KV) cache, whose size grows linearly with sequence length and dominates memory and bandwidth at deployment (Zhang et al., 2023; Xiao et al., 2023; Li et al., 2024). A large literature reduces this cost through *eviction*, a binary retain-versus-remove decision over cached tokens (Zhang et al., 2023; Xiao et al., 2023; Li et al., 2024; Tang et al., 2024; Cai et al., 2024; Guo et al., 2024; An et al., 2026). More recent work exploits *key/value asymmetry*, treating the two cache components differently under quantization or low-rank projection (Liu et al., 2024; Tao et al., 2024; Li et al., 2025; Saxena et al., 2024; Sun et al., 2024a; Jiang and Wang, 2026; Lin et al., 2026). These methods repeatedly find that keys are more sensitive than values (Liu et al., 2024; Tao et al., 2024; Li et al., 2025), inviting an appealing intermediate state between “keep” and “drop.”

For softmax attention, a cached entry i contributes $\exp(z_i)v_i$ to the numerator and $\exp(z_i)$ to the shared denominator $Z = \sum_j \exp(z_j)$. Zeroing v_i while keeping its key removes only the numerator term; dropping the entry also removes $\exp(z_i)$ and renormalizes the survivors. The former stores the exact key at half the ordinary per-entry byte cost. This creates a natural three-way decision: FULL (key and value), ADDRESS (key only, zero value), or DROP (neither). It also poses a concrete deployment question:

Outside forced prefix and recent windows, do modern instruction-tuned RoPE decoders contain a distributed, byte-substantial class of entries whose exact keys remain useful after their values are zeroed?

An affirmative answer would justify an address-only cache primitive alongside eviction and quantization. We answer this question through a pre-registered, optimizer-robust, multi-document audit and obtain a *bounded negative*. We deliberately favor the address-only hypothesis: an oracle observes exact per-coalition reconstruction error under the model’s output projection W_O , chooses jointly among all three states with grouped-query-attention (GQA) semantics, and is checked against exhaustive enumeration by a nonlinear optimizer. A weak optimizer therefore cannot explain the result.

Across Qwen2.5-3B, Qwen2.5-7B (Yang et al., 2024), and Mistral-7B (Jiang et al., 2023), on three disjoint PG-19 documents (Rae et al., 2019) at 2K, 4K, and 8K endpoints, the strongest allocator selects only **1.3415%** of eligible entries as ADDRESS, worth **0.6708%** corrected byte saving. Every model, document, and endpoint fails the preregistered 20% ADDRESS and 15% saving gates. Pointwise search does find 50 gate-passing strategy rows over 26 of 440 physical model/layer/head locations, including a reproducible Mistral layer-0 pocket that recurs across all documents, endpoints, and 12 eligibility-window settings. An exact-layer-matched permutation test rejects the intuitive “sink-adjacent” explanation. Powered downstream equivalence tests on RULER (Hsieh et al., 2024) further show that a matched-byte binary cache is equivalent to the ternary cache within a prespecified ± 1 -point accuracy margin for all three models.

The paper contributes an optimizer-robust auditing framework with exact enumeration, validated nonlinear search, a GQA-aware byte accountant, and an independent verifier. It then establishes an aggregate negative, characterizes rather than hides the sparse exceptions, and identifies functional co-movement between value and denominator importance outside those pockets. Finally, it illustrates three practical evaluation pitfalls: an apparent negative can be caused by a weak allocator, an apparent sink association can be caused by layer confounding, and a nonsignificant downstream difference is not evidence of equivalence. We address these pitfalls through exhaustive validation, exact-layer matching, and two-one-sided equivalence testing. We do *not* claim pointwise absence, generalization beyond standard RoPE decoders at contexts up to 8K, or general inferiority of asymmetric quantization and low-rank compres-

sion.

2 Related Work

Work on KV-cache efficiency falls into three mechanism-level families. Our contribution is positioned against all three.

KV eviction. Eviction methods retain a subset of cached tokens and discard the rest. H2O keeps recent tokens plus accumulated-attention “heavy hitters” (Zhang et al., 2023); StreamingLLM keeps initial sink tokens and a recent window (Xiao et al., 2023); SnapKV clusters important prompt tokens using an observation window (Li et al., 2024); Quest performs query-aware page selection (Tang et al., 2024); PyramidKV assigns layer-varying budgets (Cai et al., 2024); VATP incorporates value norms into importance (Guo et al., 2024); and ReST-KV and OB-Cache score removal through output reconstruction (An et al., 2026; Gu et al., 2025). These methods make a binary decision. Our ExactBundle-Binary control instantiates that regime exactly, while ReST-min operationalizes single-deletion reconstruction ranking as a matched-byte baseline. We ask whether a third state adds measurable value.

Key/value asymmetry. KIVI quantizes keys per channel and values per token (Liu et al., 2024); AsymKV uses layer-wise asymmetric precision (Tao et al., 2024); KVTuner tunes mixed precision and analyzes greater key sensitivity (Li et al., 2025); Eigen Attention and ShadowKV exploit low-rank structure and offloading (Saxena et al., 2024; Sun et al., 2024a); Sub-Token Routing retains selected value groups while leaving keys intact (Jiang and Wang, 2026); and VECTOR routes entries among retention, approximation, and eviction (Lin et al., 2026). VECTOR is closest to our three-way framing. These studies establish that keys and values can benefit from different representations; they do not imply that an exact-key/zero-value state forms a distributed class. Our result bounds that specific interpretation without contradicting gains from quantization, approximation, or low-rank recovery.

Attention sinks and non-informative destinations. StreamingLLM identifies initial-token sinks (Xiao et al., 2023); Massive Activations describes input-independent implicit-bias activations that concentrate attention (Sun et al., 2024b);

Gu et al. propose a key-bias account with non-informative value computation (Gu et al., 2024); and DuoAttention distinguishes retrieval from streaming heads (Xiao et al., 2024). These accounts predict that address-only-favorable entries may be sink-adjacent. We test this prediction directly and reject it after controlling exactly for layer. Downstream equivalence uses the two-one-sided-tests (TOST) procedure (Lakens, 2017), avoiding the common error of interpreting failure to reject a difference as evidence for equivalence. We evaluate on RULER (Hsieh et al., 2024), which broadens needle retrieval with tracing, aggregation, and variable-needle tasks. A complementary benchmarking effort quantifies what long-context capabilities must be sacrificed under KV compression (Yuan et al., 2024); our study contributes a targeted negative for one specific intermediate cache state rather than a broad ranking of compression families.

3 Problem Setup and Decision Rule

We study one query position attending over past entries in a RoPE decoder (Su et al., 2021) with GQA (Ainslie et al., 2023). Let g denote a KV-head group and $\mathcal{H}(g)$ its query heads. For query head h , dense attention is

$$o_h^* = \frac{\sum_{i=1}^T \exp(z_{hi}) v_i}{\sum_{i=1}^T \exp(z_{hi})}. \quad (1)$$

An assignment $s_i \in \{\text{FULL}, \text{ADDRESS}, \text{DROP}\}$ determines a denominator mask $m(s_i)$ and a value mask $u(s_i)$:

s_i	FULL	ADDRESS	DROP
$m(s_i)$	1	1	0
$u(s_i)$	1	0	0
$c(s_i)$	2	1	0

where c counts one key-sized byte slot and one value-sized byte slot. The intervened output is

$$o_h(s) = \frac{\sum_i m(s_i) \exp(z_{hi}) u(s_i) v_i}{\sum_i m(s_i) \exp(z_{hi})}. \quad (2)$$

Thus ADDRESS preserves the exact key logit and denominator term but contributes a zero message; DROP removes both and renormalizes.

For group g , we compare the dense and intervened outputs after concatenation and the corresponding columns of the output projection:

$$\mathcal{L}_g(s) = \left\| W_O^{(g)} (\text{concat}_{h \in \mathcal{H}(g)} o_h^* - \text{concat}_{h \in \mathcal{H}(g)} o_h(s)) \right\|_2^2. \quad (3)$$

The oracle minimizes Equation (3) subject to $\sum_i c(s_i) \leq B_g$. Prefix and recent windows are forced to FULL to protect sinks and locality, following common eviction practice (Xiao et al., 2023; Li et al., 2024). The accountant partitions bytes per KV head, preventing a group with more associated query heads from silently receiving more storage.

Preregistered aggregate decision. An eligible entry is address-only-favorable only when the joint optimizer assigns it to ADDRESS under the matched-byte objective. The aggregate hypothesis passes only if both the ADDRESS fraction is at least 20% and corrected byte saving is at least 15%. If A of n eligible entries use ADDRESS, then the saving relative to full BF16 KV over those entries is $A/(2n)$ because ADDRESS saves one of two equal-sized slots. We evaluate the conjunction both in the pooled audit and separately by model, document, and endpoint. Pointwise gate-passing rows are analyzed separately and cannot overturn an aggregate NO-GO; this distinction prevents a small but reproducible pocket from being misreported as a distributed cache class.

4 Method: An Optimizer-Robust Audit

Exact coalition objective. For the exact sweep, each GQA group contains $n = 8$ eligible entries. We enumerate all $3^8 = 6,561$ ternary coalitions and all $2^8 = 256$ binary coalitions. Every candidate is evaluated with the shared denominator in Equation (2) and the W_O -projected loss in Equation (3); replacing this objective with independent token scores would miss interactions induced by renormalization. This design choice is important: because ADDRESS and DROP share the same numerator term (both zero the value message) but differ only in whether they retain $\exp(z_{hi})$ in the denominator, their relative benefit is intrinsically coupled through the softmax normalizer and cannot be scored token-by-token without approximation error.

Allocators. In addition to exhaustive enumeration, we evaluate an additive singleton surrogate and a multistart nonlinear local search over ternary assignments. The surrogate estimates marginal ADDRESS and DROP costs per entry and allocates under a byte budget. The nonlinear search is the strongest optimizer reference. It matched the exhaustive optimum on all 15,840 audited strategy

rows, so the aggregate negative is not attributable to local-search failure within the audited $n = 8$ setting. The role of the additive surrogate is to enable larger allocation sweeps where full enumeration is infeasible; we never rely on it for the headline claim, and its approximation gap relative to the exact optimum is what motivates reporting the exhaustive-agreement statistic prominently.

Intervention and verifier checks. An all-FULL assignment must reproduce dense attention, while ADDRESS must preserve its denominator contribution exactly. Independent reconstruction checked 48 saved assignments across all nine extraction files and found maximum discrepancy 4.857×10^{-17} . The verifier recomputes coalition losses, byte totals, aggregate and pointwise gate decisions, sink statistics, and downstream tests. The complete suite passes 93 of 93 checks.

5 Experimental Setup

Models and source documents. We test three instruction-tuned RoPE decoders with GQA: Qwen2.5-3B, Qwen2.5-7B (Yang et al., 2024), and Mistral-7B (Jiang et al., 2023). Fresh activations are regenerated from three disjoint PG-19 documents (Rae et al., 2019). Nine pinned 8K extraction files, one per model–document pair, are reused at query endpoints 2K, 4K, and 8K. Every file passes manifest validation, has zero projected-attention reconstruction discrepancy at the recorded precision, and contains no nonfinite tensor. Llama-3.1-8B was inaccessible under the available gate and is out of scope; no tested model exceeds 8B parameters and no context exceeds 8K tokens.

Controls. Full BF16 KV is the unmodified reference. ExactBundle-Binary, the primary scientific control, permits only FULL and DROP under identical forced windows and byte partitions. AddressOnlyKV adds ADDRESS. AddressOnly-Random preserves per-group state counts but permutes token identities over five seeds. ReST-min ranks binary retention by exact single-deletion projected-output error at matched budget (An et al., 2026; Gu et al., 2025). SubToken-min (Jiang and Wang, 2026) is restricted to our Walsh–Hadamard implementation and is reported only as implementation-specific context, not as a general comparison to asymmetric compression.

Slice	ADDRESS (%)	Save (%)	Gates	Decision
Pooled	1.3415	0.6708	fail/fail	NO-GO
Qwen2.5-3B	1.5336	0.7668	fail/fail	NO-GO
Qwen2.5-7B	1.7299	0.8650	fail/fail	NO-GO
Mistral-7B	1.1176	0.5588	fail/fail	NO-GO
Document 0	1.2879	0.6440	fail/fail	NO-GO
Document 1	1.3542	0.6771	fail/fail	NO-GO
Document 2	1.3826	0.6913	fail/fail	NO-GO
2K endpoint	1.0748	0.5374	fail/fail	NO-GO
4K endpoint	1.3210	0.6605	fail/fail	NO-GO
8K endpoint	1.6288	0.8144	fail/fail	NO-GO

Table 1: Aggregate address-only audit. ADDRESS is the percentage of eligible non-prefix entries assigned to ADDRESS by the strongest allocator. Save is the corrected $A/(2n)$ byte saving. The preregistered gates are 20% ADDRESS and 15% saving; every slice fails both.

Downstream evaluation. We compare matched-byte binary and ternary caches on RULER (Hsieh et al., 2024). Accuracy differences use task-stratified paired bootstrap intervals. Equivalence is assessed by TOST with a prespecified ± 1 -percentage-point margin (Lakens, 2017) and retrospective power. This tests whether the complete confidence region lies within a practically negligible range; merely obtaining a nonsignificant ordinary difference test would not support the same claim.

6 Results

6.1 The aggregate result is a clean NO-GO

Across 15,840 strategy rows and 126,720 exact $n = 8$ entries, the strongest allocator selects 1,700 ADDRESS assignments: **1.3415%**, worth **0.6708%** corrected byte saving. Both values are far below the preregistered gates (Table 1). Every model is individually NO-GO: Qwen2.5-3B reaches 1.5336% ADDRESS and 0.7668% saving, Qwen2.5-7B reaches 1.7299% and 0.8650%, and Mistral-7B reaches 1.1176% and 0.5588%. Document-level estimates are tightly clustered. The endpoint trend is monotone, increasing from 1.0748% at 2K to 1.3210% at 4K and 1.6288% at 8K, but even the 8K estimate remains more than twelvefold below the 20% gate. Within the tested range, longer context does not approach byte-substantiality.

The monotone endpoint trend deserves emphasis because it is the one direction in which the hypothesis could plausibly have strengthened. As context grows, more entries fall outside the forced

recent window, so a genuinely distributed address-only class would have a larger pool in which to appear and its prevalence should rise toward the gate. The prevalence does rise – by roughly 0.55 percentage points from 2K to 8K – but it rises within a regime that is still an order of magnitude below the threshold. Linear extrapolation of this slope would not reach 20% until context lengths far beyond any we can currently validate, and such extrapolation is explicitly disclaimed by our scope. The practical reading is that the address-only state does not become byte-substantial merely because there are more old tokens to place in it.

6.2 Pointwise absence is false

The aggregate negative does not imply that the phenomenon occurs nowhere. A pointwise search over 440 physical model/layer/head locations finds 50 gate-passing strategy rows spread over 26 locations. The dominant structure is a Mistral-7B layer-0 pocket: KV heads 1–4 produce 24 passing records across all three documents and all three endpoints. The pocket recurs under all 12 tested eligibility-window settings, with L0/H3 recurring in 12 of 12 settings, while tested sibling heads never pass. The exceptions are therefore reproducible rather than optimizer noise, but they remain sparse and concentrated in early layers. This pattern is compatible with prior evidence that lower layers require distinct cache treatment (Cai et al., 2024) and that KV requirements vary by head (Xiao et al., 2024); it is not sufficient to justify a distributed address-only primitive.

The scale asymmetry between the two findings is the crux of the paper. Twenty-six passing locations out of 440 is a 5.9% location-level hit rate, but each location contributes only a handful of the tens of thousands of eligible entries pooled into the aggregate. When the passing entries are counted against the full eligible population rather than against the location roster, they are absorbed into the 1.3415% pooled figure. A study that reported only the pointwise search – a natural choice for an author hoping to motivate a new primitive – would legitimately observe reproducible, model-specific pockets and could frame them as evidence for an address-only cache class. Our two-level reporting shows why that framing would be misleading at the granularity that matters for a systems allocator, which must decide bytes over the whole cache rather than over a hand-picked head.

6.3 Exact layer matching rejects sink adjacency

The attention-sink literature (Xiao et al., 2023; Sun et al., 2024b; Gu et al., 2024) predicts that address-only-favorable locations may be non-informative attention destinations. We preregistered a permutation test to determine whether the “sink-adjacent” label should be retained. The exact-layer analysis contains 3,960 matched units and 41 positive units, using 200,000 permutations within model, document, endpoint, and layer. Neither metric is enriched after Holm correction: the conventional sink metric C4 has pass-minus-control difference -81.2 with Holm $p = 1.0$, and the secondary-sink metric S1 has difference $+0.021$ with Holm $p = 1.0$. A stored-attention cross-check has maximum absolute error zero.

This negative is methodologically informative. A looser layer-quartile match appears significant (Holm $p = 3.5 \times 10^{-5}$), but the signal disappears when controls come from the same exact layer. The apparent association is therefore explained by the concentration of both variables in early layers rather than sink overlap. Concretely, both address-only-favorable pockets and conventional sink activations are disproportionately early-layer phenomena, so any matching scheme coarse enough to pool early and late layers into the same stratum will attribute their shared early-layer excess to a spurious overlap. Only when the control set is drawn from the identical layer does the confound dissolve, and it dissolves completely. We therefore reject “sink-adjacent” and use the narrower description *sparse reproducible early-layer pockets*. This is a transferable warning: mechanistic labels that ride on depth should be validated against depth-matched, not depth-quartile-matched, controls.

6.4 Value and denominator importance co-move

For every eligible entry we compute

$$\text{adv} = \frac{\text{drop_err}}{\text{addr_err}}.$$

If keeping a key without its value were broadly useful, dropping the entry would be much worse than choosing ADDRESS, producing $\text{adv} \gg 1$. Across 17.6 million eligible entries, the per-model medians are 0.9874 for Qwen2.5-3B, 1.0029 for Qwen2.5-7B, and 1.0075 for Mistral-7B. Address

Matched-byte comparison	Median error ratio
AddressOnly / Binary	1.000
AddressOnly / ReST-min	1.000

Table 2: Reconstruction-error ratios over 140 groups at 62.5% byte budget. A ratio of 1.000 indicates median parity; adding the ADDRESS state does not improve over the binary controls.

and drop therefore have nearly equal reconstruction error outside the sparse pockets. We describe this as *functional inseparability*, not as a causal account: the value message and denominator contribution co-move under our intervention, so retaining only the latter rarely buys useful accuracy. The result also bounds a universal key-norm explanation (Gu et al., 2024) and agrees with evidence that value information matters for eviction (Guo et al., 2024).

Three of the four medians sit slightly above one and one sits slightly below, which is exactly the signature of a near-degenerate choice: for the typical entry the allocator is nearly indifferent between paying one byte to preserve the denominator and paying zero bytes to drop the entry outright, because the denominator perturbation that DROP introduces is small relative to the value message that both states already discard. When the denominator term is large enough that its removal would hurt – the case where ADDRESS should win decisively – the entry usually also carries a value message worth keeping, so the optimizer prefers FULL. The intermediate state is squeezed from both sides. This is a mechanistic reason, grounded in the shared softmax normalizer, for why the aggregate prevalence is small rather than merely an empirical coincidence.

6.5 Matched-byte controls show parity

In a 140-group audit at 62.5% budget, the median AddressOnly/Binary and AddressOnly/ReST-min reconstruction-error ratios are both exactly 1.000 (Table 2). The extra state therefore provides no median improvement over either exact binary allocation or reconstruction-ranked binary retention. AddressOnly-Random separately checks that retaining the same numbers of FULL, ADDRESS, and DROP entries at arbitrary identities does not constitute the scientific effect. SubToken-min remains implementation-specific context and is not used to support a general ordering among compression families.

Model	A–B (points)	Power	± 1 equiv.
Qwen2.5-3B	−0.424	0.928	yes
Qwen2.5-7B	−0.137	0.923	yes
Mistral-7B	−0.196	0.997	yes

Table 3: RULER equivalence of AddressOnly (A) and matched-byte Binary (B). “Yes” means both one-sided tests reject at the prespecified ± 1 -percentage-point margin.

The matched-byte comparison is the decisive systems statement. A budget-unconstrained comparison would trivially favor the ternary policy, because a strictly larger action set can never increase reconstruction error. The relevant question for a deployment engineer is instead whether the extra state helps *at equal cost*, and here the median improvement is exactly zero against both an exact binary optimizer and a reconstruction-ranked binary heuristic that mirrors deployed eviction scorers. That the tie holds against ReST-min in particular matters, because ReST-min is not an oracle: it is the kind of single-deletion importance ranking a practitioner would actually implement, and it already captures whatever the address-only state could have contributed.

6.6 Downstream equivalence holds for all models

On RULER, ternary and binary caches are equivalent within the prespecified ± 1 -point margin for all three models (Table 3). Mistral-7B is particularly informative because it contains the strongest exception pocket. At $n = 511$ (73 examples per task, zero execution errors, and zero score-recomputation mismatches), AddressOnly minus Binary is -0.196 points with 90% CI $[-0.587, +0.098]$ and power 0.9965. Qwen2.5-7B was increased to $n = 1,092$ (156 examples per task) with a byte-identical ordered-prefix check for the original records; its difference is -0.137 points with 90% CI $[-0.641, +0.321]$ and power 0.923. Qwen2.5-3B has difference -0.424 points and power 0.928. Both one-sided component tests reject their respective nonequivalence nulls for every model. Thus, even where a reproducible pocket exists, the ternary policy has no practically meaningful downstream advantage over matched-byte binary caching at the stated margin.

Notably, all three point estimates are mildly negative, meaning the ternary cache is if anything

slightly *worse* than the matched-byte binary cache on RULER, though never by a practically meaningful or statistically distinguishable amount. This is consistent with the reconstruction-level parity: with no median accuracy headroom at the tensor level, the extra state cannot translate into downstream gains, and small negative point estimates are the expected outcome of a policy that spends part of its byte budget on entries whose value message has been zeroed. We stress that the equivalence claim rests on the confidence region lying inside ± 1 point, not on the difference being non-significant; the achieved power of 0.92–0.997 is what licenses reading these as equivalence results rather than as underpowered nulls.

6.7 Optimizer and eligibility robustness

Two checks bound alternative explanations. First, multistart nonlinear search matches exhaustive enumeration on all 15,840 strategy rows; a weak optimizer cannot account for the low ADDRESS prevalence. Second, a multi-model sweep over 12 prefix/recent-window settings remains aggregate NO-GO in every setting. The most permissive pooled result reaches 3.47% ADDRESS and 1.74% saving, still below both gates, while the Mistral L0/H3 pocket recurs in all 12 settings and tested sibling heads never pass. The aggregate negative and the localized exception therefore survive changes to the eligibility carve-out. Because the forced prefix and recent windows are the most obvious lever an advocate could pull to inflate prevalence – shrinking them exposes more old tokens to the optimizer – their inability to breach the gate across a twelvefold configuration sweep is strong evidence that the negative is not an artifact of a conservative eligibility choice.

7 Practical Lessons

The audit exposes several pitfalls relevant to real deployment studies. First, an appealing algebraic distinction need not produce an allocatable systems primitive. Preserving the denominator appears useful in isolation, but byte-matched joint optimization shows that binary retention obtains the same median reconstruction quality. Second, sparse exceptions can make both extreme narratives wrong: claiming universal absence would ignore the reproducible Mistral pocket, while promoting that pocket as a general cache class would ignore its 1.3415% pooled prevalence. Reporting

both aggregation levels is essential.

Third, mechanistic labels require confounder-matched tests. The sink interpretation would have survived a layer-quartile comparison, yet fails under exact-layer matching; the broader label would therefore be misleading. Finally, downstream “no difference” must be supported by an equivalence design rather than an underpowered null-hypothesis test. The ± 1 -point TOST analysis establishes a bounded practical statement, whereas an ordinary nonsignificant comparison would establish only insufficient evidence of a difference. These checks turn a negative result into actionable guidance: for the tested RoPE decoders and contexts, engineering effort is better directed toward binary selection, quantization, or structured approximation than toward a general exact-key/zero-value cache state.

A fifth lesson concerns the value of favoring the hypothesis one hopes to refute. Every discretionary modeling choice in this study – an oracle with access to exact projected reconstruction error, joint rather than greedy allocation, exhaustive enumeration as ground truth, and a permissive eligibility sweep – was made to give the address-only state its best possible chance. The negative therefore carries more weight than a negative produced by a weak baseline, because the obvious rebuttal (“a stronger optimizer would have found the class”) is closed off by construction. We recommend this adversarial-to-oneself posture for negative-result deployment studies generally: the credibility of a NO-GO is largely determined by how hard the design tried to reach GO.

8 Conclusion

We asked whether modern RoPE decoders contain a distributed, byte-substantial non-prefix address-only KV class. Under an exact-reconstruction surrogate verified against exhaustive optimization, the answer is a bounded negative: 1.3415% ADDRESS and 0.6708% saving in aggregate, every model/document/endpoint NO-GO, and powered ± 1 -point downstream equivalence for all three models. Pointwise absence is false: 50 gate-passing rows span 26 of 440 locations, including a reproducible Mistral layer-0 pocket. Exact-layer matching rejects sink adjacency, and drop/address error ratios near one identify functional inseparability elsewhere. We release the framework, allocators, byte accountant, exception and sink maps,

downstream provenance, and verifier. Future work should increase the document-cluster count, test additional architectures, and probe contexts of at least 16K tokens, streaming dialogue, and long-reasoning workloads.

Limitations

This result is a bounded negative, not a claim of pointwise absence. Only three PG-19 documents were completed rather than the proposed six to eight, limiting document-cluster uncertainty despite stable per-document aggregates of 1.29%, 1.35%, and 1.38%. The additive singleton surrogate is approximate; the central claim instead relies on nonlinear local search matched to exhaustive enumeration for all 15,840 audited rows, but a different objective could identify other pockets. We test three models from two families; Llama-3.1-8B was gated, and the conclusions do not cover ALiBi, NoPE, mixture-of-experts, or models above 8B parameters. Contexts do not exceed 8K, so extrapolation to 16K or longer contexts, streaming generation, multi-turn chat, and long-reasoning traces is unsupported. Exact enumeration uses groups of eight eligible entries, bounding optimization error within that audit but not proving global optimality for arbitrary cache-length coalitions. Finally, SubToken-min uses a restricted Walsh–Hadamard implementation and supplies only implementation-specific context, not a general asymmetric-compression comparison.

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A Audit Configuration

Table 4 collects every fixed configuration reported in the study. The main aggregate sweep contains 15,840 strategy rows and 126,720 exact entries. Each row has $n = 8$ eligible entries, making exhaustive evaluation of all 6,561 ternary and

256 binary assignments feasible. The nine 8K files correspond to three models and three disjoint documents; 2K and 4K endpoints are evaluated from the same pinned extraction family rather than counted as separate extraction files.

Component	Configuration
Models	Qwen2.5-3B, Qwen2.5-7B, Mistral-7B
Source	3 disjoint PG-19 documents
Endpoints	2K, 4K, 8K
Pinned extraction files	9
Exact eligible entries/group	8
Ternary coalitions/group	6,561
Binary coalitions/group	256
Strategy rows	15,840
Exact entries	126,720
Aggregate gates	20% ADDRESS, 15% saving
Matched-byte audit	140 groups, 62.5% budget
Random-control seeds	5
Sink permutations	200,000
TOST equivalence margin	± 1 percentage point
Verifier checks	93/93 passed

Table 4: Reported audit configuration. No unreported model, context length, or training run is included.

B Objective and Allocation Details

The selected-coalition objective is evaluated per GQA group. ADDRESS sets v_i to zero while retaining the exact post-RoPE key logit in the shared softmax denominator. DROP removes both the numerator and denominator contributions and renormalizes the surviving entries. FULL retains both. Forced prefix and recent entries remain FULL and are excluded from address-only prevalence. The byte accountant assigns cost two to FULL, one to ADDRESS, and zero to DROP, then partitions the budget by KV head.

The additive singleton surrogate computes an independent marginal cost for each candidate state. This approximation is useful for larger allocation sweeps but cannot represent all softmax-renormalization interactions. The multistart nonlinear search directly evaluates the nonlinear coalition objective and serves as the strongest allocator. Exhaustive agreement on 15,840 of 15,840 rows is the relevant optimizer-validity result; the paper does not claim that the additive surrogate itself is exact.

The corrected saving is reported separately from prevalence. If A among n eligible entries are assigned ADDRESS, they save A value slots relative to $2n$ full-KV slots, yielding $A/(2n)$. Reporting ADDRESS prevalence itself as byte saving would

therefore overstate the systems benefit by a factor of two.

C Extraction and Numerical Validation

Nine pinned 8K extraction files were regenerated from three model checkpoints and three disjoint PG-19 documents. Manifest validation found no nonfinite tensors. The recorded projected-attention reconstruction discrepancy was zero at the stored precision. The independent verifier then reconstructed 48 saved assignments distributed across all nine files; its maximum discrepancy was 4.857×10^{-17} . An all-FULL assignment reproduces the dense model, and stored-attention cross-checks used by the sink analysis have maximum absolute error zero.

The previous raw activation-generation state and original RULER execution state were not persisted. We therefore distinguish regenerated model activations from read-only recovery of downstream records. Regenerated extractions are pinned by manifests. Recovered RULER records are checked through score recomputation and nested ordered-prefix provenance before additional examples are appended.

D Complete Reported Aggregate Results

Table 5 repeats the complete pooled and one-dimensional slice results in a single appendix table. The endpoint increase is monotone, whereas variation across documents is small. Because only three document clusters are available, the latter stability should not be interpreted as a precise population-level uncertainty estimate.

Slice	ADDRESS count/rate	Saving	Decision
Pooled	1,700/126,720 (1.3415%)	0.6708%	NO- GO
Qwen2.5-3B	1.5336%	0.7668%	NO- GO
Qwen2.5-7B	1.7299%	0.8650%	NO- GO
Mistral-7B	1.1176%	0.5588%	NO- GO
Document 0	1.2879%	0.6440%	NO- GO
Document 1	1.3542%	0.6771%	NO- GO
Document 2	1.3826%	0.6913%	NO- GO
2K endpoint	1.0748%	0.5374%	NO- GO
4K endpoint	1.3210%	0.6605%	NO- GO
8K endpoint	1.6288%	0.8144%	NO- GO

Table 5: All reported pooled and one-dimensional aggregate slices. Both gates must pass; every slice fails both.

E Exception Map

The pointwise audit covers 440 physical model/layer/head locations and produces 50 gate-passing strategy rows over 26 locations. The strongest coherent pocket occurs in Mistral-7B layer 0. KV heads 1–4 account for 24 passing records across all three documents and all three endpoints. The eligibility-window sweep contains 12 prefix/recent-window settings; L0/H3 recurs in all 12, while tested sibling heads never pass. These observations support reproducibility of the local exception, not prevalence of a global class.

The 41 positive units in the exact-layer sink analysis are not the same counting object as the 50 passing strategy rows. The latter are raw strategy-level records; the former are units retained after constructing the exact-layer-matched permutation analysis. Keeping these denominators separate prevents the permutation sample from being mistaken for the physical-location prevalence estimate.

F Sink-Overlap Procedure

The conventional sink family follows the position-0/prefix-attention account in StreamingLLM (Xiao et al., 2023). The secondary-sink family follows the high-attention, low-information-message account motivated by massive activations (Sun et al., 2024b). Permutations are restricted within the tuple (model, document, endpoint, layer), producing 3,960 matched units with 41 positives. We use 200,000 permutations and apply Holm correction across the designated metrics.

The conventional C4 statistic has pass-minus-control difference -81.2 and Holm-adjusted $p = 1.0$. The secondary S1 statistic has difference -0.021 and Holm-adjusted $p = 1.0$. In contrast, matching only by layer quartile gives Holm-adjusted $p = 3.5 \times 10^{-5}$. Because the apparent effect disappears under the stricter design, the pre-registered “sink-adjacent” label is rejected. The quartile result is retained as a confounding diagnostic rather than presented as evidence for the mechanism.

G Co-Movement and Norm Diagnostics

Across 17.6 million eligible entries,

$$\text{adv} = \text{drop_err} / \text{addr_err}$$

has model medians 0.987357 for Qwen2.5-3B, 1.00285 for Qwen2.5-7B, and 1.00754 for Mistral-7B. Values close to one mean that preserving the denominator while zeroing the value has nearly the same reconstruction cost as removing the entry. This supports functional inseparability outside the sparse pockets. It does not prove that keys and values have identical causal roles, nor does it contradict gains obtained by representing them at different quantization precisions.

Per-model key/value-norm Spearman analyses were treated as secondary diagnostics. They do not yield a universal norm mechanism capable of explaining the distributed hypothesis. Accordingly, the main text states only the supported negative conclusion and does not infer a common causal origin from norm correlations.

H Matched-Byte Controls

The matched-byte audit evaluates 140 groups at a 62.5% budget. Median reconstruction-error ratios are AddressOnly/Binary = 1.000 and AddressOnly/ReST-min = 1.000. ExactBundle-Binary uses the same forced windows, per-KV-head byte accounting, and projected-output objective as AddressOnlyKV, differing only in the absence of ADDRESS. ReST-min ranks entries using exact single-deletion projected-output discrepancy. AddressOnly-Random preserves each group’s numbers of FULL, ADDRESS, and DROP assignments and permutes their token identities over five seeds.

SubToken-min is deliberately scoped more narrowly. The implementation uses a restricted Walsh–Hadamard construction and is not treated as a complete reproduction of every sub-token routing design. It therefore supplies implementation-specific context but is excluded from broad claims about whether token-level or sub-token-level asymmetry is superior.

I Downstream Equivalence and Provenance

RULER (Hsieh et al., 2024) accuracy is compared between matched-byte binary and ternary caches using task-stratified paired bootstrap intervals and ± 1 -percentage-point TOST equivalence tests (Lakens, 2017). Mistral-7B completion recovered five missing qa_2 examples through isolated resume, reaching $n = 511$ (73 examples per task) with zero execution errors and zero score-

recomputation mismatches. Its AddressOnly-minus-Binary estimate is -0.196 points, with 90% CI $[-0.587, +0.098]$ and power 0.9965.

Qwen2.5-7B was increased from $n = 511$ to $n = 1,092$ (156 examples per task). A byte-identical ordered-prefix check verifies the provenance of the original records before the top-up. Its difference is -0.137 points, with 90% CI $[-0.641, +0.321]$ and power 0.923. Qwen2.5-3B has difference -0.424 points and power 0.928. Both one-sided tests reject their respective nonequivalence nulls for every model. Had a model failed to reach the target power within the available budget, we would have reported the achieved power rather than asserting equivalence.

J Eligibility-Window Ablation

The eligibility ablation evaluates 12 prefix/recent-window configurations across all three models. Every configuration remains aggregate NO-GO. The most permissive pooled configuration reaches 3.47% ADDRESS and 1.74% corrected saving, still below the 20% and 15% gates. At the same time, the Mistral-7B L0/H3 location recurs in 12 of 12 settings while tested sibling heads never pass. The ablation therefore supports both sides of the bounded conclusion: the global class remains insubstantial, and the local pocket remains reproducible.

K Compute

The framework, nine extractions, full exception sweep, co-movement analysis, matched-byte audit, and Qwen2.5-3B downstream run were completed and reused. Subsequent targeted compute consisted of approximately 0.5 A100-day for sink-overlap testing on stored statistics and approximately 2–3 A100-days for Mistral-7B RULER completion and the Qwen2.5-7B power top-up, totaling approximately 3–4 A100-80GB-days. The study performs no training, uses no model above 8B parameters, evaluates no context above 8K tokens, and uses no proprietary API.

L Release and Verification Artifacts

The release contains the reference formulas, tiled solver, additive and nonlinear allocators, byte accountant, test suite, extraction manifests, exception maps, sink-overlap maps, prevalence and advantage statistics, clustered-analysis scripts, down-

1002 stream outputs with provenance and checksums,
1003 and a verifier that recomputes coalition losses,
1004 byte totals, gate decisions, permutation statistics,
1005 and bootstrap/TOST results. Large extraction ar-
1006 rays are represented by checksummed manifests
1007 rather than duplicated copies. The verifier test
1008 suite reports 93 of 93 passing checks.